RUNI – Algorithms – Homework 2

Due date: Sunday 16/11/2025

NOTE: the general homework guidelines published in homework 1 are valid for all homework assignments in this course.

Problem 1:

a. Let G=(V,E) be a **connected undirected** graph.

Prove that it is possible to add at most $\lfloor n/2 \rfloor$ edges to E such that the resulting graph has an Euler cycle.

b. Let G=(V,E) be a **directed** graph such that for every $v \in V$, $d_{in}(v) = d_{out}(v)$

Prove that G is strongly connected if and only if its underlying graph is connected.

a.

As shown in the recitation, the number of vertices with an odd degree in G is even. Thus, the maximum number of vertices having odd degree is n if n is even, or n-1 if n is odd. Partition the vertices with odd degrees into at most $\lfloor n/2 \rfloor$ pairs, and connect every pair with a new edge. The resulting graph is with all even degree vertices, hence has an Euler cycle.

b.

It was shown in the recitation that: a strongly connected graph has an Euler cycle if and only if for every $v \in V$, $d_{in}(v) = d_{out}(v)$.

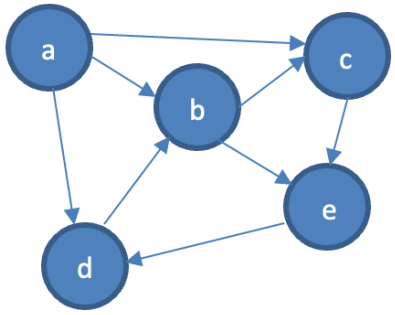
Similarly, it was shown in class that: a directed graph has an Euler cycle if and only if its underlying graph is connected and for every $v \in V$, $d_{in}(v) = d_{out}(v)$.

If G is strongly connected then G has an Euler cycle and the first direction follows from the class claim. If the underlying graph of G is connected then G has an Euler cycle and the second direction follows from the recitation claim.

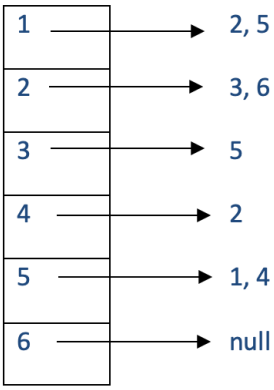
Problem 2:

For each of the following graphs find and present (separately) an Euler path and a Hamilton path, or claim that they do not exist (no proof required).

a. The graph provided in the following figure:



b. The graph provided in the following representation:



c. The graph provided in the following representation:

	a	b	c	d	e	f
a	0	1	0	0	0	0
b	0	0	1	0	1	1
c	0	0	0	0	1	1
d	0	0	1	0	0	0
e	0	2	0	0	0	0
f	0	0	0	1	0	0

a. Euler path: does not exist
Hamilton path: a, c, e, d, b

b. Euler path: 1,2,3,5,1,5,4,2,6

Hamilton path: does not exist

c. Euler path: a,b,c,f,d,c,e,b,e,b,f

Hamilton path: a,b,f,d,c,e

Problem 3:

In each of the following questions, select the correct answer and justify. No need to refute the wrong answers.

3.1 The sequence 00120221 is a De-Bruijn **sub**sequence over $\Sigma = \{0,1,2\}$ and some n.

a. It must be that $n = 2$.
b. If the next character is 0, then it must be that $n > 2$.
c. There is no Euler cycle in the De-Bruijn graph $G_{3,2}$.

3.2 Let G be a graph in which every Hamiltonian cycle is also an Euler cycle.

a. G must be a clique

b. G must be a cycle

c. G is not necessarily a clique or a cycle.

3.3 Let $G = (V_1, V_2, E)$ be an **undirected bipartite** graph **with an Euler cycle**.

a. $|V_1| = |V_2|$
b. $|V_1|$ and $|V_2|$ are even.
c. $\sum_{u \in V_1} \deg(u) = \sum_{v \in V_2} \deg(v)$

3.1. The correct answer is b.

There are $3^2 = 9$ words of length 2 over $\Sigma = \{0,1,2\}$. Given the next character, the subsequence is already of length 9 so represents 9 words. The word 00 repeats in index 0 and index 8 (cyclicly).

3.3. The correct answer is b. G must be a cycle.

A cycle is the only graph in which all vertices have degree 2.

3.3. The correct answer is c. $\sum_{u \in V_1} \deg(u) = \sum_{v \in V_2} \deg(v)$

This is true in every bipartite (independent of the existence of an Euler cycle). Every edge increases by 1 the sum of degrees of each side (part) of the bipartite.

Problem 4:

Let G=(V,E) be an **undirected connected** graph, such that **all its vertices have even degrees**.

A vertex *s* is called an Euler-center if every application of the algorithm for finding an Euler cycle in G, starting at *s*, produces an Euler cycle in a single iteration, no matter what is the order in which edges are selected. In other words, there is no need to merge different parts of the cycle into others. Complete the proof of the following theorem:

If *s* is an Euler-center then *s* appears in every simple cycle in G.

Proof: Assume that *s* is an Euler-center.

Assume towards contradiction that s does not appear in some cycle C. Let G\C be the graph G with the edges of C removed. Formally, V(G\C)=V, E(G\C)=E\C.

(complete the proof)

G\C is a graph of even degrees, so the connected component of G\C that includes s has an Euler cycle. Let C' be an Euler cycle of G\C that starts and ends with s. Consider the application of the algorithm on G, that starts with the cycle C'. Since the degree of s in G\C' is 0 (as s does not appear in C), then this application of the algorithm needs more than one iteration, contradicting the assumption that s is an Euler-center.

Problem 5:

In a recent debacle, thieves disguised as construction workers stole eight pieces of the French Crown Jewels valued at approximately €88 million from the Galerie d'Apollon of the Louvre in Paris, France. Lucas Philippe, head of the French interpol requested your help in analyzing the patrol route of the night of the incident.

The head guard at the Louvre shared the following details:

- There are seven major areas: Apollo (A), Denon (D), Egyptian (E), Greek (G), Napoléon (N), Richelieu (R), and Sully (S).
- There is one bidirectional (undirected) corridor between every pair of areas.
- A single guard performs a continuous patrol walking each corridor exactly once.

a. Will a patrol route that uses every corridor exactly once and returns to its starting area be possible under these conditions? Give a brief yes/no justification.

b. On occasion, corridors may close for renovation. Given a subset E' of open corridors, propose a method to efficiently decide whether a continous patrol that uses all corridors exactly once exists.

c. After much investigation, it was found that on the day of grand burglary the area Sully (S) was entirely closed, along with all corridors connected to it. The night guard reports he managed to adjust his patrol and had performed a route that uses every corridor exactly once, and starts and ends in the Apollo area. Is the guard telling the truth? Give a brief yes/no justification.

a. Yes. Model the museum as a complete undirected graph K_7.

Each area (vertex) has degree 6 (even), and the graph is connected, so an Euler cycle exists.

b. Construct the graph G' = (V,E').

Check that: (1) G' is connected; (2) either all vertices have even degree (Euler cycle) or exactly two have odd degree (Euler path).

c. No. Removing S and its six corridors leaves K_6.

Each remaining area then has degree 5 (odd), giving 6 odd-degree vertices. No Euler cycle possible.